

etc. The good dictionaries may be a few GBs in size and can be used to hack the most common passwords. This method won't work if the network's password isn't on the word list. With a dictionary of decent size, the attack takes a reasonable amount of time.

iii. Rainbow table

The validation process against the 4-way handshake involves hashing the password and comparing it with the hash in a handshake. Hashing is a CPU intensive task and is the bottleneck in the whole process. Some tools use GPUs instead of CPUs to speed up the process. You might be thinking whether it would be better if the dictionaries contained hashes instead of plaintext passwords. Well, no. WPA-2 takes the network's name (SSID) into account while hashing. So, the wordlist has to contain separate hashes for all possible SSIDs. Although not possible for all SSIDs, it is feasible to generate hashes for all the default SSIDs of common routers. Precomputed hashes significantly reduce the time taken to crack a network (if the network SSID was present in the word list). The table of hashes is known as the rainbow table. They are much larger than the wordlists (as they contain several hashes of the same plaintext password). As we improve the time complexity, we compromise on space complexity. This is known as the time-memory tradeoff and is a major dilemma in the entire computer industry. Few rainbow tables can be as large as 100 GBs.

How to carry out the attack

Carrying out this attack used to be difficult at an earlier point in time. Now, it's a breeze. If you have all the prerequisites installed, then hacking the network is as simple as:

- `reaver -i <interface-name> -b <BSSID of target>`

This process may take a few hours. Let your machine be and check after ten minutes or so, and the progress bar must be at least 1% by now. Use this time for your leisure or tag along and feel like a hacker sitting in front of the screen flashing code.

Information Gathering

Now before we run the above script, we need to check a few things-

- Is WPS enabled or not. If not, then the attack will most probably not work.